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BASIC RESEARCH ARTICLE

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Mental health during and after the COVID-19 pandemic – a longitudinal study over 42 months in five European countries

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ABSTRACT

Background: The mental health impact of the COVID-19 pandemic is well documented. However, only a few studies investigated mental health in later phases of the pandemic and after its official end. Moreover, little is known about people's psychological burden related to the pandemic and other global crises post-pandemic.

Objective: Study's first objective was to compare mental health outcomes in the general population over the course of the pandemic and ten months post-pandemic. The second objective was to explore people's psychological burden regarding the pandemic, in comparison to current wars, climate crises, inflation, and poor government management and/or corruption in the post-pandemic era.

Method: Participants from the general population of Austria, Croatia, Germany, Greece, and Portugal (68.8% female, $M_{age} = 41.55$) were assessed online up to four times between June 2020 and March 2024 (baseline sample: $N = 7913$). *Adjustment Disorder New Module – 8* (ADNM-8), *Patient Health Questionnaire* (PHQ-2), and *World Health Organization-Five Well-Being Index* (WHO-5) were used to measure adjustment disorder, depression, and well-being. Prevalence rates were calculated and repeated measures ANOVAs applied to assess mental health at four time points. One-way repeated measures ANOVA was run to explore how the different global crises were related to participants' burden.

Results: Temporal variations in mental health were evident across four assessment waves, with highest levels of probable adjustment disorder and depression in winter 2020/2021 (T2). A slight improvement of mental health was found at later time points. Current wars and inflation were the greatest sources of psychological burden at the post-pandemic assessment, revealing some cross-country differences.

Conclusion: Although mental health differences in the general population were not as pronounced as in the acute phase of the pandemic, psychosocial support is still needed post-pandemic. This is likely to be due to other global crises that take a toll on people's mental health.

La salud mental durante y después de la pandemia de COVID-19: un estudio longitudinal de 42 meses en cinco países europeos

Antecedentes: El impacto de la pandemia de COVID-19 en la salud mental está bien documentado. Sin embargo, solo unos pocos estudios investigaron la salud mental en las fases posteriores de la pandemia y tras su finalización oficial. Además, se sabe poco acerca de la carga psicológica de las personas en relación con la pandemia y otras crisis mundiales post-pandemia.

Objetivo: El primer objetivo del estudio fue comparar los resultados de salud mental en la población general en el curso de la pandemia y diez meses posteriores. El segundo objetivo fue explorar la carga psicológica de las personas en relación con la pandemia, en comparación a las guerras actuales, la crisis climática, la inflación y la mala gestión gubernamental y/o corrupción en la era post pandémica.

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Salud mental; trastorno adaptativo; bienestar; COVID-19; pandemia; guerra; cambio climático; inflación; corrupción

HIGHLIGHTS

- The sample included 7913 participants from Austria, Croatia, Germany, Greece, and Portugal who participated in a 4-wave online study between June 2020 and March 2024.
- Mental health outcomes changed over the course of the pandemic and after its official end, with the highest symptoms being found in times of strict preventive measures and high number of COVID-19 cases and deaths.
- The findings indicate that current wars, inflation, and financial difficulties represented a major source of psychological burden for people in Europe in early 2024.

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Método: Participantes de la población general de Austria, Croacia, Alemania, Grecia, y Portugal (68.8% mujeres, edad promedio = 41.55) fueron evaluados en línea hasta cuatro veces entre junio 2020 y marzo 2024 (muestra basal: *N* = 7.913). Se utilizaron el Nuevo Módulo 8 para el Trastorno Adaptativo (ADNM-8 en sus siglas en inglés), Cuestionario de Salud del Paciente (PHQ-2 en sus siglas en inglés), y el Índice de Bienestar de la Organización Mundial de la Salud-versión 5 (WHO-5 en sus siglas en inglés) para medir el trastorno adaptativo, la depresión y el bienestar. Se calcularon las tasas de prevalencia y se aplicaron ANOVAs de medidas repetidas para evaluar la salud mental en cuatro momentos. Se realizó un ANOVA unidireccional de medidas repetidas para explorar cómo las diferentes crisis mundiales se relacionaban con la carga de los participantes.
Resultados: Se observaron variaciones temporales en la salud mental a lo largo de las cuatro olas de evaluación, con los niveles más altos de probable trastorno adaptativo y depresión en el invierno 2020/2021 (T2). Una leve mejoría en la salud mental se encontró en los puntos posteriores. Las guerras actuales y la inflación fueron las mayores fuentes de carga psicológica en la evaluación post pandemia, revelando algunas diferencias entre países.
Conclusión: Aunque las diferencias en salud mental en la población general no fueron tan pronunciadas como en la fase aguda de la pandemia, el apoyo psicosocial sigue siendo necesario aun en la post pandemia. Esto se debe probablemente a las otras crisis mundiales que afectan negativamente la salud mental de las personas.

1. Introduction

In March 2020, the World Health Organisation (WHO) declared the global outbreak of the novel coronavirus disease (COVID-19) pandemic (WHO, 2020). Since then, several reviews and meta analyses have shown that mental health worsened during the pandemic (Bower et al., 2023; Cénat et al., 2022; Witteveen et al., 2023). The increased prevalence rates of different mental health disorders were related to disruptions in daily routine and measures to control the virus spread, such as school closure, cancellation of mass gatherings, and quarantine (Knox et al., 2022; Liu et al., 2024b). Not all individuals were affected by these lifestyle adjustments to the same extent, with studies reporting stable and low trajectories of symptoms both in general (Lu et al., 2022) and clinical samples (Rzeszutek & Gruszczyńska, 2023). Positive psychological changes and decreases in some forms of distress were also observed (Brülhart et al., 2021; Zrnić Novaković et al., 2022). In Europe, the rise of mental health problems was particularly evident early in the pandemic, followed by an improvement later in 2020 and in 2021 (Ahmed et al., 2023).

The WHO announced the end of COVID-19 as a public health emergency in May 2023, highlighting that it remains a threat to global health (E. Harris, 2023). Despite this still ongoing threat, scientific interest in the mental health impact of the pandemic seems to have faded over time. The majority of studies collected data during 2020 and 2021, with a comparably low number of studies using data collected during 2022 and 2023 (Ahmed et al., 2023; Kim, 2024). To assess the long-term impact of the pandemic on mental health, data from later phases of the pandemic and longitudinal studies with longer follow-up periods are required. Yet most studies conducted later in the pandemic were cross-sectional, reporting prevalence rates and predictors of mental health problems. According to these studies, symptoms of depression, anxiety, stress, and adjustment disorders were elevated at later stages of the pandemic (Alasqah et al., 2023; Fu & Xie, 2024; Hilal et al., 2024). The described psychological difficulties were linked to tiredness and distress due to COVID-19 preventive measures and demotivation to follow them (Abdul Rashid et al., 2023). Other factors impacting mental health later in the pandemic were female gender, financial situation, education, and history of mental health problems (Liu et al., 2024b; Takaew et al., 2024; Wirkner & Brakemeier, 2024; Wu et al., 2024).

So far, only few longitudinal and repeated cross-sectional studies were carried out in later phases of the pandemic, indicating different courses of mental health trajectories over time. According to an Italian study, perceived levels of stress, depression, and anxiety increased from April 2020 to April 2021, and remained high even in April 2022 (Della Cattaneo Volta et al., 2024). A German multiwave study conducted between December 2020 and December 2022 revealed highest depression and anxiety levels during the last wave (Wirkner & Brakemeier, 2024). However, there is also evidence of a decrease in depression and post-traumatic symptoms between 2020 and 2022, pointing towards people's capacity to adapt to the pandemic over time (Fukase et al., 2023).

When looking into mental health towards the end of the pandemic, the findings are also somewhat contradictory. Although there are some findings on amelioration of symptoms in spring 2023 compared to spring 2022 (Li et al., 2024), the majority of studies conducted shortly before the WHO declared the end of the pandemic (i.e. May 2023) suggested a high level of psychological problems both in students (Luo et al., 2024; Xin et al., 2024) and health care workers (Liu et al., 2024b; Wu et al., 2024). Yet the trajectories of mental health changes might differ depending on the target population or the disorder investigated. In people who were infected with COVID-19, anxiety and insomnia scores remained stable between spring 2022 and spring 2023, while depression scores decreased in the same period (Badinlou et al., 2024). Finally, it is possible that different individuals show different patterns of change in mental health over time, as demonstrated by an Australian study conducted between December 2020 and June 2022 (Donohoe-Bales et al., 2024).

Taken together, data about mental health in later phases of the COVID-19 pandemic and after its official end are rather scarce and inconclusive. Most of the existing studies were conducted on students, healthcare workers, and in Asian countries, limiting their generalisability. There is also a paucity of longitudinal studies with assessment waves covering the pandemic and post-pandemic phase, and the existing studies yielded inconsistent results. Furthermore, the research was mainly focused on anxiety and depression, neglecting other disorders amidst the pandemic (e.g. adjustment disorder) and indicators of positive psychological functioning (e.g. well-being). Addressing these research gaps is crucial for a better understanding of long-term psychological adjustment to the COVID-19 pandemic and post-pandemic effects. In addition, the inclusion of a post-pandemic assessment can provide insights about improvement or deterioration of mental health following the lifting of COVID-19 restrictions.

Notably, mental health after the pandemic might be impacted by other global crises at the time. For instance, the negative consequences of climate change and financial crisis on mental health have been recently discussed in the scientific literature (Di Quirico, 2023; Lawrance et al., 2022). To the best of our knowledge, only one cross-sectional study explored the interaction between the COVID-19 pandemic, climate change, ongoing wars, and financial difficulties, showing elevated negative affectivity and depressive symptoms resulting from the cumulative impact of these crises (Kałwak et al., 2024). Research on adjustment disorder might generate further knowledge on pandemic-and crisis-related distress, given that this disorder is characterised by troubling thoughts, ongoing worry, and difficulties in different areas of functioning (WHO, 2019). These symptoms are likely to develop in times of uncertainty and changes, such as major crises mentioned above.

1.1. Aims of the present study

The present study set out to address the previously indicated knowledge gaps regarding mental health and COVID-19 pandemic, considering other global crises that might impact people's mental health post-pandemic. The first aim was to examine changes in mental health over the course of the pandemic and post-pandemic across multiple European countries. To capture different psychological responses, symptoms of depression and adjustment disorder, and levels of well-being were assessed. The second aim was to explore the perceived psychological burden of people in Europe in early 2024 regarding the COVID-19 pandemic and its consequences, in comparison to other global crises (i.e. inflation and financial difficulties, poor government management and/or corruption, current wars, as well as climate change and natural disasters). Recognising people's capacity for resilience and successful adjustment following adverse life events (Bonanno, 2004), we hypothesised an improvement in mental health over time, with best outcomes at the post-pandemic assessment. At this last assessment, we further assumed that the pandemic will be perceived as less burdensome than other more salient crises, such as current wars and climate change.

2. Method

In spring 2020, the European Society for Traumatic Stress Studies (ESTSS) launched a longitudinal study (ADJUST study) to assess the mental health impact of the COVID-19 pandemic, with a special focus on adjustment disorder (study protocol: Lotzin et al., 2020; OSF preregistration: https://doi.org/10.17605/OSF.IO/8XHYG). The study included 11 European countries which conducted three assessment waves: T1 (June 2020–December 2020), T2 (December 2020–June 2021), and T3 (June 2021–January 2022). An additional fourth wave (T4) was carried out in countries having resources for the post-pandemic assessment, namely: Austria, Croatia, Greece, Germany, and Portugal, between January and March 2024, approximately 10 months after the end of the pandemic was declared by the WHO (E. Harris, 2023). The data collected in these countries are the subject of the analysis presented in this paper.

2.1. Participants and procedure

In all five countries, the four assessment waves were conducted as online surveys. Convenience sampling was used to recruit participants for the first wave. To reach a population as heterogeneous as possible, the link to the first online survey was distributed to different companies, organisations, universities, sport associations, cultural institutions (e.g. theatres, museums, libraries), and clubs for senior citizens. A detailed overview of recruitment strategies per country is provided elsewhere (Lotzin et al., 2021). People who participated in the first wave and agreed to be re-contacted for further assessments provided their e-mail address, which was used to send invitations for participation in the second, third, and forth wave, respectively.

The inclusion criteria for participation included: (1) a minimum age of 18 years, (2) currently living in Austria, Croatia, Germany, Greece, or Portugal, (3) being able to read and write in the language of the respective study site, and (4) providing informed consent. A total of $N = 7913$ ($M_{age} = 41.55$, $SD = 12.95$) participants, predominantly female ($n = 5446$, 68.8%), met the inclusion criteria at T1. Out of them, $n = 6486$ provided baseline data on all main mental health outcomes (adjustment disorder, depression, or well-being), whereas $n = 7131$ provided data on at least one outcome.

In all five participating countries, an ethical approval was obtained before the link for the first assessment wave was distributed. The details on each ethical approval are presented below:

Austria: Ethics Committee of the University of Vienna (Reference Number: 00554)
Croatia: Ethics Committee of the Department of Psychology, Faculty of Humanities and Social Sciences, University of Zagreb (Date of approval: 21/05/2020)
Greece: Social Sciences Ethics Review Board (SSERB), University of Nicosia (SSERB 00109)
Germany: Local Psychological Ethics Committee at the Centre for Psychosocial Medicine at the University Medical Center Hamburg-Eppendorf (LPEK-0149)
Portugal: Ethics Committee of the Medical Faculty, University of Porto and University Hospital Center of São João, Porto (CE 201–20).

2.2. Measures

Validated self-report questionnaires were used to assess mental health outcomes at all four assessment waves. Cronbach's Alpha or Spearman-Brown Coefficient was used as measure of internal consistency, depending on the number of items in the respective questionnaire (Eisinga et al., 2013). The descriptions of questionnaires are provided below and internal consistency at each assessment wave can be found in Table 1.

Table 1. Internal consistency of measures used in the study.

	T1: 06/2020–12/2020 (*n*)	T2: 12/2020–06/2021 (*n*)	T3: 06/2021–01/2022 (*n*)	T4: 01/2024–03/2024 (*n*)
ADNM-8	.92 (7130)	.92 (2181)	.92 (1998)	.92 (1521)
PHQ-2	.78 (6486)	.83 (2168)	.83 (1988)	.80 (1520)
WHO-5	.90 (6520)	.91 (2135)	.91 (1953)	.90 (1483)

Note. All available cases were used for the calculation of internal consistency, which is why the reported sample sizes differ across measures and timepoints. ADNM-8, Adjustment Disorder New Module-8; PHQ-2, 2-Item Patient Health Questionnaire; WHO-5, World Health Organization-Five Well-Being Index. For measuring internal consistency, Cronbach's Alpha was used for ADNM-8 and WHO-5, and Spearman-Brown Coefficient for PHQ-2.

The *Adjustment Disorder New Module – 8* (ADNM-8; Kazlauskas et al., 2018) is a brief measure of adjustment disorder symptoms following the criteria of the 11th version of the International Classification of Diseases. Good psychometric properties of the ADNM-8 have been shown in multiple studies on different populations (Ben-Ezra et al., 2018; B. E. Harris et al., 2023). The measure was successfully used in COVID-19 research to assess pandemic-related adjustment difficulties (e.g. Bambrah et al., 2022).

The *Patient Health Questionnaire-2* (PHQ-2; Kroenke et al., 2003) is an ultra-brief screening scale for depression. There is a large body of evidence on clinical utility and diagnostic accuracy of the PHQ-2 (Hlynsson & Carlbring, 2024; Staples et al., 2019). This screening tool was often administered to measure depressive symptoms during the COVID-19 pandemic (e.g. Hettich et al., 2022).

The *World Health Organization-Five Well-Being Index* (WHO-5; WHO, 1998) is a well-established instrument for measuring well-being. It has been used in different settings, with numerous studies reporting its good psychometric properties (Lara-Cabrera et al., 2022; Topp et al., 2015). The WHO-5 was also frequently used in the context of the COVID-19 pandemic (e.g. Würtzen et al., 2022).

In addition to these mental health outcomes, at T4 (post-pandemic) we also assessed self-reported burden due to five global crises: (i) COVID-19 pandemic and its consequences, (ii) inflation and financial difficulties, (iii) poor government management and/or corruption, (iv) current wars, and (v) climate change and natural disasters. The participants were asked to rate how burdensome they find each of these crises. The question was conceptualised as a point allocation question, where participants could distribute 10 points across the five crises, with a higher number of points indicating a greater psychological burden. This question was included to account for the variety of stressors that might impact mental health after the official end of the COVID-19 pandemic.

2.3. Data analysis

As a first step, detailed descriptive and missing data analyses were conducted. To address the first aim, missing data on mental health outcomes (ADNM-8, PHQ-2, WHO-5) were imputed using multiple imputation. Specifically, Multiple Imputation by Fully Conditional Specification, also referred to as Multiple Imputation by Chained Equations (White et al., 2011) was applied. As proposed in the literature, inclusive strategy was adopted, by using multiple auxiliary variables (i.e. covariates with high correlations with the variable to be imputed) in the imputation model (Collins et al., 2001; Graham, 2009). According to Hippel (2020), a two-way procedure was followed to calculate the number of required imputations. In the first stage, a pilot analysis was conducted with 20 imputations. Using this pilot dataset, a repeated-measures ANOVA was conducted to estimate the fraction of missing information (FIM). The average FIM was 78.7%, ranging between 73.9% and 84.8% across the four time points. Thus, 80 imputations were deemed sufficient to generate the final multiply imputed dataset which was used in further analyses on mental health outcomes.

Self-reported prevalence rates of adjustment disorder and depression were calculated based on previously established cut-off scores of ≥23 for ADNM-8 (Kazlauskas et al., 2022; Zelviene et al., 2020) and ≥3 for PHQ-2 (Kroenke et al., 2003). Cochran's *Q* test was then run to determine if the prevalence rates were different at four time points, followed by pairwise comparisons using Dunn's procedure with a Bonferroni correction. Next, one-way repeated measures ANOVA was run for each mental health outcome to examine whether the levels of adjustment disorder, depression, and well-being differed at four time points. In cases where Mauchly's test indicated the violation of sphericity assumption, the Greenhouse–Geisser correction was used. For all ANOVAs, a post-hoc analysis with a Bonferroni adjustment was conducted. Given the large body of evidence on gender differences in mental health amidst COVID-19, we further explored mental health outcomes in men and women across four time points using *t*-tests. Finally, all analyses were re-run using only data without missing values (i.e. complete case analysis with original data) as a sensitivity analysis, to check whether the results based on these two methods were comparable.

To address the second aim, one-way repeated measures ANOVA was used to test whether participants reported different levels of burden for the different crises. Squared root transformation was applied on the data due to violation of normality and a significant number of outliers, while the Greenhouse–Geisser correction was applied due to violation of sphericity assumption. Finally, Kruskal–Wallis test was run five times for each of the five crises to explore whether participants in different countries indicated different levels of burden.

The descriptive analyses, analyses regarding the second aim, and multiple imputation were conducted in SPSS (version 28.0; IBM Corp, 2021), whereas all analyses with multiply imputed data were conducted in R (version 2024.04.2; R Core Team, 2022), using the following packages: *haven* (version 2.5.4), *afex* (version 1.4.1), and *tidyr* (version 1.3.1).

3. Results

3.1. Participants' characteristics

At T1, most participants were from Croatia (29.9%) and Germany (34.5%). More than a half (62.3%) had a university degree and were full-time employed (59.9%). Whereas 38.7% reported having high income, 20.9% reported having low or very low income. A large proportion was in a relationship (67.3%) and had one or more children (46.2%). One-third of the sample described their health as 'very good' (34.1%), and almost one-fifth had a mental health diagnosis at some point in their life (17.9%). Detailed information on the participants' baseline characteristics per country, along with participant flow and results of the missing data analysis can be found in the Supplement S1.

3.2. Aim 1: mental health at four assessment waves

3.2.1. Prevalence rates of mental health outcomes

As shown in Table 2, the self-reported prevalence rates of adjustment disorder and depression were significantly different at four time points. Post-hoc tests indicated that the prevalence rate of adjustment disorder at T2 was higher than at other three time points ($p < .001$). Similarly, the prevalence rate of depression at T2 was higher than the prevalence rates at T1 ($p = .002$), T3 and T4 (both $p < .001$).

3.2.2. Mean scores of mental health outcomes

Figures 1–3 show mental health outcomes at four time points for the original data and multiply imputed data. The ANOVA results on temporal changes in mental health outcomes are summarised in Table 3. Post-hoc analysis revealed that adjustment disorder and depression symptoms increased significantly between T1 and T2, then decreased significantly between T2 and T3 ($p < .001$). A further decrease between T3 and T4 was significant only for adjustment disorder ($p < .001$). Well-being levels significantly decreased between T1 and T2 ($p < .001$), followed by a significant increase between T2 and T3 ($p < .001$), and a further stabilisation at T4.

Table 2. Differences in self-reported prevalence rates of probable adjustment disorder and depression at four time points

		n (%)	χ^2	*df*	*p*
Adjustment disorder (*N* = 1003)					
T1:	06/2020–12/2020	151 (15.1)	48.96	3	<.001
T2:	12/2020–06/2021	213 (21.2)			
T3:	06/2021–01/2022	141 (14.1)			
T4:	01/2024–03/2024	131 (13.1)			
Depression (*N* = 1000)					
T1:	06/2020–12/2020	182 (18.2)	39.16	3	<.001
T2:	12/2020–06/2021	233 (23.3)			
T3:	06/2021–01/2022	164 (16.4)			
T4:	01/2024–03/2024	149 (14.9)			

Note. Only participants who responded to all four measures were included in this analysis (i.e. complete case analysis without imputation).

Table 3. Differences in levels of adjustment disorder, depression, and well-being at four time points – Results of One-Way Repeated Measures ANOVA.

	df between groups	*df* within groups	*F*-statistic	*p*-value	η^2_G	*MSE*
Pooled statistics for ANOVA run on multiply imputed data						
ADNM-8	2.86	22622.13	333.19	< .001	.02	17.61
PHQ-2	2.95	23319.13	221.43	< .001	.01	1.13
WHO-5	2.96	23448.84	309.12	< .001	.01	194.40
ANOVA statistics for original data (i.e. complete case analysis)						
ADNM-8	2.77	2775.92	45.80	< .001	.03	*Only computed in case of imputation*
PHQ-2	2.91	2908.67	25.83	< .001	.02	
WHO-5	2.94	2843.35	40.84	< .001	.03	

Note. MSE, Mean Squared Error (i.e. within-imputation variance); η^2_G, Generalised Eta Squared; ADNM-8, Adjustment Disorder New Module-8; PHQ-2, 2-Item Patient Health Questionnaire; WHO-5, World Health Organization-Five Well-Being Index.

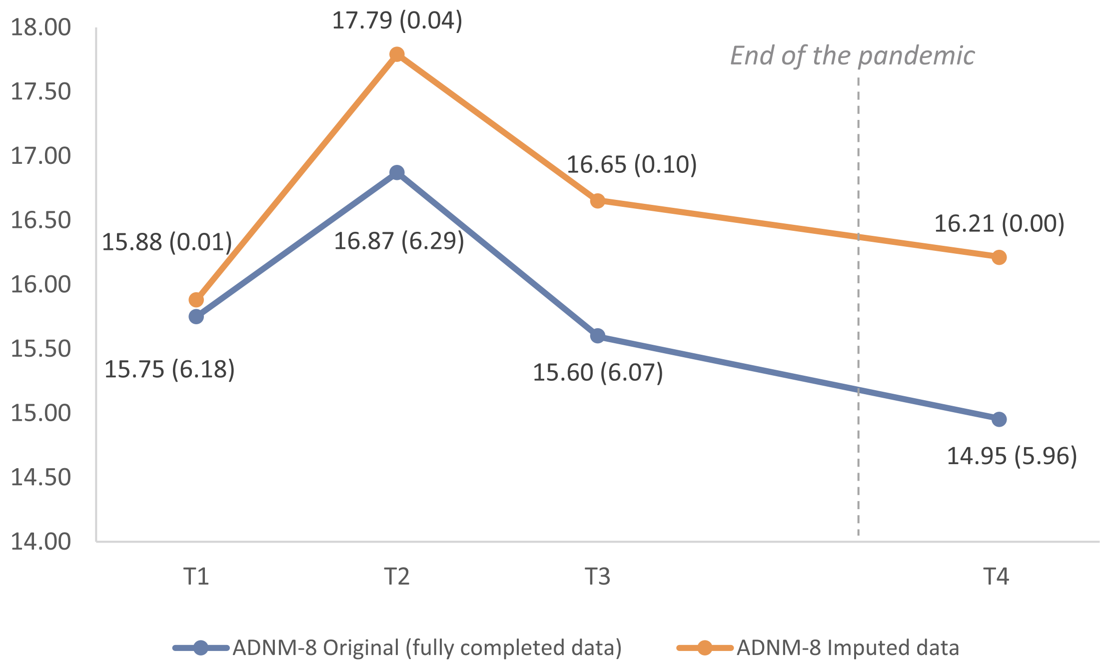

Figure 1. Symptoms of adjustment disorder (*M*, *SD*) at four time points.
Note: T1: 06/2020–12/2020; T2: 12/2020–06/2021; T3: 06/2021–01/2022; T4: 01/2024–03/2024. ADNM-8, Adjustment Disorder New Module-8.

3.2.3. Gender differences

The analysis with the imputed data revealed significantly better mental health outcomes in men compared to women at all assessment waves. According to the sensitivity analysis with original data, the difference was not significant only for depressive symptoms at T1 and T2. Detailed results are shown in the Supplement S1.

3.3. Aim 2: psychological burden due to major crises

The participants' burden was differently pronounced depending on the respective crisis, $F(3.51, 4927.77) = 300.73$, $p < .001$, partial $\eta^2 = .18$. As shown in Table 4, current wars represented the major psychological burden in the overall sample. However, the descriptive analyses suggested that other sources of burden were predominant in other countries, e.g. inflation and financial difficulties in Greece.

The Kruskal–Wallis test confirmed that there were significant differences in the level of psychological burden regarding the five major crises between the participating countries. According to the calculated effect sizes, the largest differences between the five countries were found with respect to inflation and financial difficulties, followed by current wars and climate change. For details, see Table 5.

4. Discussion

The present study pursued two key objectives: to assess changes in mental health across Europe during and after the COVID-19 pandemic, and to explore people's psychological burden regarding the pandemic and other major crises in early 2024 (i.e. post-pandemic). We found significant variations in depression, adjustment disorder, and well-being at four time points, with worst mental health outcomes in winter 2020/2021 (T2). At the post-pandemic assessment, self-reported prevalence rates of depression and adjustment disorder decreased. At this assessment, current wars and financial differences represented the major psychological burden for the

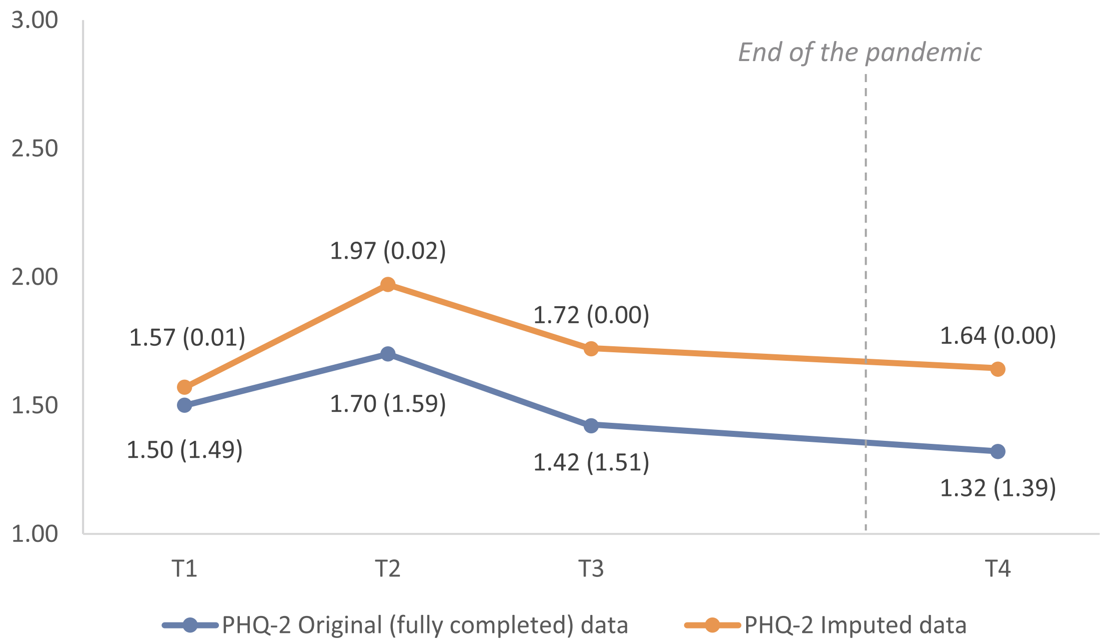

Figure 2. Symptoms of depression (*M*, *SD*) at four time points.

Note: T1: 06/2020–12/2020; T2: 12/2020–06/2021; T3: 06/2021–01/2022; T4: 01/2024–03/2024. PHQ-2, 2-Item Patient Health Questionnaire.

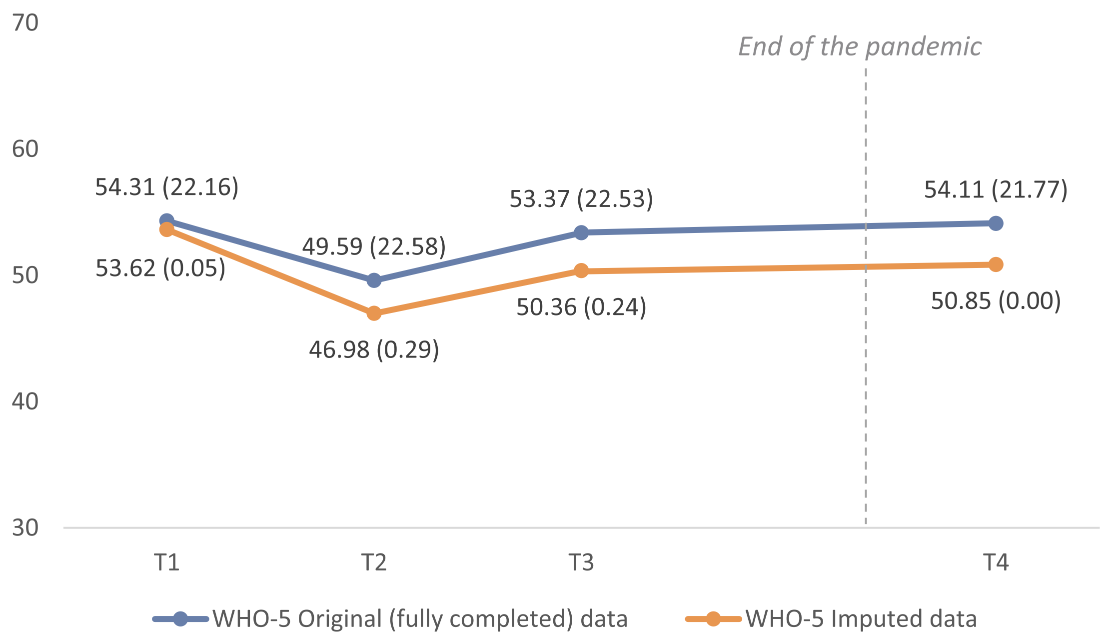

Figure 3. Levels of well-being (*M*, *SD*) at four time points.

Note: T1: 06/2020–12/2020; T2: 12/2020–06/2021; T3: 06/2021–01/2022; T4: 01/2024–03/2024. WHO-5, World Health Organization-Five Well-Being Index.

Table 4. Participants' psychological burden due to major crises.

	Overall sample	Austria	Croatia	Germany	Greece	Portugal
	n	*n*	*n*	*n*	*n*	*n*
	1406	250	358	668	86	44
	M (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)
COVID-19 pandemic and its consequences	0.94 (1.16)	1.14 (1.25)	0.84 (0.97)	0.85 (1.15)	1.27 (1.27)	1.32 (1.41)
Inflation and financial difficulties	2.25 (1.64)	1.82 (1.44)	2.99 (1.65)	1.93 (1.56)	2.88 (1.45)	2.20 (1.68)
Poor government management and/or corruption	1.98 (1.47)	1.87 (1.37)	2.27 (1.49)	1.84 (1.51)	2.14 (1.41)	1.98 (1.15)
Current wars	2.59 (1.50)	2.91 (1.55)	2.11 (1.36)	2.82 (1.51)	1.84 (1.17)	2.70 (1.37)
Climate change and natural disasters	2.24 (1.45)	2.27 (1.41)	1.78 (1.30)	2.56 (1.50)	1.87 (1.24)	1.80 (1.05)

Note. Due to the violation of several assumptions, root squared transformation was applied and the transformed data were used to conduct one-way repeated measures ANOVA. Following transformed values, i.e. *M* (*SD*) for the five crises, were used to conduct ANOVA: 0.70 (0.67), 1.35 (0.66), 1.25 (0.64), 1.51 (0.55), and 1.37 (0.61).

Table 5. Cross-country differences in the level of psychological burden for each of the five crises – results of Kruskal–Wallis test.

	df	*H*-statistic (χ^2)	*p*	η^2
COVID-19 pandemic and its consequences	4	26.07	<.001	0.02
Inflation and financial difficulties	4	152.59	<.001	0.11
Poor government management and/ or corruption	4	27.22	<.001	0.02
Current wars	4	89.35	<.001	0.06
Climate change and natural disasters	4	80.03	<.001	0.05

Note: η^2 = Eta Squared.

participants, with several cross-country differences being observed.

4.1. Mental health at four assessment waves

In winter 2020/2021 (T2), one-fifth of the participants were at increased risk of developing depressive or adjustment disorders. At this time point, levels of well-being were the lowest, highlighting considerable psychological difficulties in the general population, which is also supported by highest mean scores for depression and anxiety disorder indicated at T2. This assessment took place during the first winter of the pandemic period. In all participating countries, this winter was marked by strict measures to control the virus spread (Mathieu et al., 2020). Infection and death rates were also particularly high at T2 (WHO, 2024). Both stringency of preventive measures and pandemic characteristics have been linked to psychological problems (Knox et al., 2022; Witteveen et al., 2023) and might account for the high mental health burden found in our study during the second assessment.

The prevalence rates gradually decreased after T2, reaching the lowest values at the post-pandemic assessment. The improvement of mental health following the end of the pandemic might suggest that the gradual 'return to normality' helped the participants recover. Numerous restrictions, disrupted daily routines, fear of infection, work and financial difficulties were just some of the stressors impacting people's mental health over almost three years of the pandemic. It is plausible that the reduction or even disappearance of the pandemic-related stressors has led to mental health improvement, as shown in some studies (Li et al., 2024). However, considerable psychological difficulties after the pandemic were also reported (Badinlou et al., 2024; D. Liu et al., 2024a). This discrepancy could be attributed to individual differences in terms of psychological adjustment, which were shown during the pandemic among various study groups (Kenntemich et al., 2024; Lu et al., 2022; Rzeszutek & Gruszczyńska, 2023). Due to the methods used, subgroups with different symptom trajectories could not be investigated in the current study.

Interestingly, the comparison of results based on imputed data and those based on original data (i.e. complete case analysis) revealed some inconsistencies. Whereas both analyses suggested highest symptoms of depression and anxiety disorder at T2, the time point with lowest symptoms differed. For imputed data, lowest scores were found at the first assessment, and for the original data at the last assessment. Moreover, when looking at the original data, the increase of symptoms between T1 and T2 was less sharp and the symptom reduction faster, compared with the imputed data. This might suggest that people with better mental health were more likely to respond to more than one assessment. On the other hand, those who participated only at T1 and whose data had to be imputed for all the remaining time points, might have had more psychological difficulties, which potentially could have prevented them from further participation. This bias among longitudinal survey respondents was reported in earlier COVID-research (Czeisler et al., 2021; O'Connor et al., 2021).

In our study, temporal variations in well-being were less pronounced compared to adjustment disorder and depressive symptoms. Moreover, well-being scores followed comparable trajectories when using imputed and original data. Well-being might thus be a more robust indicator of mental health amidst pandemics. The performance of WHO-5 in longitudinal studies was rarely examined, with available studies focusing on shorter periods between assessments (Yang et al., 2023). Earlier COVID-studies also showed little variations in WHO-5 scores over time (Pieh et al., 2021). This suggests that well-being might be less sensitive to changes in mental health during a pandemic or might depict an overall mood over a longer period, which distinguishes this construct from other mental health indicators. Further investigations are required to better understand temporal changes in WHO-5.

At all time points, women reported lower well-being and higher symptoms of depression and adjustment disorder than men. There is extensive literature pointing to greater psychological difficulties in females during the COVID-19 pandemic (Witteveen et al., 2023). Our study corroborates these findings but also provides some novel insights. Namely, the effect sizes for the tested gender differences were largest at T4, suggesting a higher gender gap in mental health post-pandemic. During COVID-19, women faced a greater increase in housework and childcare responsibilities and were at higher risk of job loss compared to men (Peck, 2020). According to the conservation of resources theory (Hobfoll, 1989), such resource losses can lead to further losses, which may be reflected in larger effect sizes at the post-pandemic assessment. As a consequence of the disproportional impact of

COVID-19 on women, their mental health burden might have continued to rise after the pandemic, whereas men might have had better chances to recover. A systematic investigation of this assumption is recommended to prevent further loss spirals and safeguard women's mental health.

4.2. Psychological burden due to major crises

Apart from the COVID-19 pandemic and its long-term consequences, other global crises also have the potential to contribute to mental ill-health. The co-occurrence of multiple global crises that pose a threat to the global community can be subsumed under the term *polycrisis* (Lawrence et al., 2024; Lawrence et al., 2022). The interactions of such crises are complex and represent a greater risk than the individual impact of each. Acknowledging that several major crises might impact mental health of people in Europe following the official end of the pandemic, the second aim of our study was to explore people's psychological burden regarding: COVID-19 pandemic and its consequences, inflation and financial difficulties, poor government management and/or corruption, current wars, as well as climate change and natural disasters.

In the overall sample and in the Austrian, German, and Portuguese subsample, the current wars represented the major burden among the five crises. Negative psychological implications of the Russo-Ukrainian war on the European population have already been shown (Scharbert et al., 2024), and might be further exacerbated by the war in the Gaza Strip. Our findings support this hypothesis and highlight the need to address war-related concerns and foster psychological resilience across Europe.

In Croatia and Greece, inflation and financial difficulties were perceived as the most stressful, followed by poor government management and/or corruption. Corruption and poverty often go hand in hand, and Croatia and Greece were countries with the lowest GDP per capita in our sample (World Bank, 2025). The COVID-19 pandemic triggered economic uncertainty, with job losses and inflation impacting people's mental health (Di Quirico, 2023), whereas previous studies established a link between corruption perceptions and psychological distress (Anyan et al., 2024). Given that neither financial hardship nor corruption can be easily solved, it is essential to implement low-threshold psychosocial support to help people cope with these adverse life conditions, especially in lower-income countries.

In our sample, the participants of higher-income countries, namely Austria and Germany, described climate change and natural disasters as their second major concern. The impact of the climate crisis on mental health is well documented (Gianfredi et al., 2024; Lawrance, Thompson, et al., 2022). As outlined in a recent position paper, Europewide psychological prevention and intervention programmes, along with further research and better education, are needed to buffer the detrimental effect of climate change on mental health (Brandt et al., 2024).

In all countries, COVID-19 pandemic and its consequences were rated as the least stressful among the five crises. This was anticipated, given that sources of participants' psychological burden were assessed in early 2024, while the official end of the pandemic was declared in spring 2023 and most restrictions had already been lifted before. Interestingly, even during the acute phase of COVID-19, higher distress was attributed to climate change compared to the pandemic (Lawrance, Jennings, et al., 2022).

Taken together, it seems plausible that psychological burden due to different crises depends on participants' characteristics and socio-economic situation in the investigated country. For instance, the pandemic and its consequences might still be perceived as a major burden by people with Long-COVID (Badinlou et al., 2024). On the other hand, the climate crisis might be a more salient stressor in higher-income countries, where financial difficulties and poor government management do not impact people's daily lives to the same extent as in lower-income countries (Thompson et al., 2024). Also, the theory of shattered assumptions (Janoff-Bulman, 1992) might contribute to a deeper understanding. The more crises shatter the world, the more challenging it becomes to maintain our assumptions about benevolence and meaningfulness of the world, but also about our own self-worth. Consequently, the world might be perceived as a trembling place, in which planning and acting become more and more difficult, resulting in heightened rates of mental health disturbances. This might be specifically true for people in higher-income countries with generally more positive assumptions of the world, which might be disturbed to a greater extent following wars or natural disasters. This could account for different concern priority perspectives found in our study. Additional research is required to test this assumption.

4.3. Strengths and limitations

To the best of our knowledge, the present study is the first to investigate different mental health outcomes during and after the COVID-19 pandemic across multiple European countries. It provides an overview of mental health trajectories over a longer period, which is rarely seen in COVID-research. Apart from depression and well-being, we also assessed adjustment disorder – a frequently diagnosed, but still understudied disorder (Morgan et al., 2022). Moreover, we explored people's psychological burden due to current wars, climate change, and other major crises, providing new insights into factors that might impact mental health post-pandemic. The large sample size and the comparison of results based on fully completed and imputed data represent further strengths of our work.

Convenience sampling is an important limitation of the present study. The generalisability of the findings is further limited by the fact that participants from some countries (e.g. Germany) were overrepresented in the overall sample. Due to limited resources, we were unable to collect data on mental health in 2022 and 2023, which would have contributed to a more pronounced understanding of mental health trajectories. In addition, the time intervals between the assessment waves were rather long. Hence, we cannot draw conclusions about changes in mental health in between the assessments. Although initially planned, symptoms of anxiety were not included in the analysis, since these were not measured at all time points in Greece. Furthermore, we did not compare mental health outcomes between the countries since this was beyond the scope of our study. Cross-country comparisons with equal sample sizes per country need to be conducted to gain a better understanding of country-specific factors which shape people's mental health post-pandemic. A more detailed examination of the impact of current wars and other global crises on mental health is another promising avenue of research.

4.4. Conclusion

The COVID-19 pandemic has taken a great toll on the mental health of the European population. Our findings indicate that psychological difficulties were particularly pronounced in winter 2020/2021, when a sharp spike in COVID-19 cases and deaths was seen around Europe. The mental health indicators at other time points during and after the pandemic did not differ substantially, suggesting that psychosocial support is needed even in the post-pandemic era. This is likely to be due to other major crises (e.g. inflation, wars) faced by the European and worldwide population. Despite its limitations, the present study lays the groundwork for future research on the mid- and long-term mental health impact of the COVID-19 pandemic and other global crises.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are available on request from the corresponding author (IZN). The data are not publicly available since the study is part of a large-scale project involving multiple countries with different privacy restrictions. The R codes used for the analysis are publicly available in the following repository: OSF | Mental health during and after the COVID-19 pandemic.

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References

Abdul Rashid, M. R., Syed Mohamad, S. N., Tajjudin, A. I. A., Roslan, N., Jaffar, A., Mohideen, F. B. S., Addnan, F. H., Baharom, N., & Ithnin, M. (2023). COVID-19 pandemic fatigue and its sociodemographic, mental health status, and perceived causes: A cross-sectional study nearing the transition to an endemic phase in Malaysia. *International Journal of Environmental Research and Public Health*, *20*(5), 4476. https://doi.org/10.3390/ijerph20054476

Ahmed, N., Barnett, P., Greenburgh, A., Pemovska, T., Stefanidou, T., Lyons, N., Ikhtabi, S., Talwar, S., Francis, E. R., Harris, S. M., Shah, P., Machin, K., Jeffreys, S., Mitchell, L., Lynch, C., Foye, U., Schlief, M., Appleton, R., Saunders, K. R. K., … Johnson, S. (2023). Mental health in Europe during the COVID-19 pandemic: A systematic review. *The Lancet Psychiatry*, *10*(7), 537–556. https://doi.org/10.1016/S2215-0366(23)00113-X

Alasqah, I., Alhamdan, A., Alhouri, M., & Alfehaid, M. (2024). Sociodemographic determinants of depression, anxiety, and stress immediately after the COVID-19 pandemic: A cross-sectional study among university students in Saudi arabia. *Frontiers in Public Health*, *11*, 1271536. https://doi.org/10.3389/fpubh.2023.1271536

Anyan, F., Andoh-Arthur, J., Adjei, S. B., & Akotia, C. S. (2024). Mental health problems, interpersonal trust, and socio-cultural correlates of corruption perception in Ghana. *Frontiers in Public Health*, *12*, 1269579. https://doi.org/10.3389/fpubh.2024.1269579

Badinlou, F., Rahimian, F., Hedman-Lagerlöf, M., Lundgren, T., Abzhandadze, T., & Jansson-Fröjmark, M. (2024). Trajectories of mental health outcomes following COVID-19 infection: A prospective longitudinal study. *BMC Public Health*, *24*(1), 452. https://doi.org/10.1186/s12889-024-17997-x

Bambrah, V., Wyman, A., Friedman, E., & Eastwood, J. D. (2022). Examining the longitudinal associations between adjustment disorder symptoms and boredom during COVID-19. *Behavioral Sciences*, *12*(9), 311. https://doi.org/10.3390/bs12090311

Ben-Ezra, M., Mahat-Shamir, M., Lorenz, L., Lavenda, O., & Maercker, A. (2018). Screening of adjustment disorder: Scale based on the ICD-11 and the adjustment disorder new module. *Journal of Psychiatric Research*, *103*, 91–96. https://doi.org/10.1016/j.jpsychires.2018.05.011

Bonanno, G. A. (2004). Loss, trauma, and human resilience: Have we underestimated the human capacity to thrive after extremely aversive events? *American Psychologist*, *59*(1), 20–28. https://doi.org/10.1037/0003-066x.59.1.20

Bower, M., Smout, S., Donohoe-Bales, A., O'Dean, S., Teesson, L., Boyle, J., Lim, D., Nguyen, A., Calear, A. L., Batterham, P. J., Gournay, K., & Teesson, M. (2023). A hidden pandemic? An umbrella review of global evidence on mental health in the time of COVID-19. *Frontiers in Psychiatry*, *14*, 1107560. https://doi.org/10.3389/fpsyt.2023.1107560

Brandt, L., Adorjan, K., Catthoor, K., Chkonia, E., Falkai, P., Fiorillo, A., Gondek, T. M., Le Vay, J. N., Rojnic, M., Meyer-Lindenberg, A., Heinz, A., Dom, G., & Luykx, J. J. (2024). Climate change and mental health: Position paper of the European psychiatric association. *European Psychiatry*, *67*(1), e41. https://doi.org/10.1192/j.eurpsy.2024.1754

Brülhart, M., Klotzbücher, V., Lalive, R., & Reich, S. K. (2021). Mental health concerns during the COVID-19 pandemic as revealed by helpline calls. *Nature*, *600*(7887), 121–126. https://doi.org/10.1038/s41586-021-04099-6

Cénat, J. M., Farahi, S. M. M. M., Dalexis, R. D., Darius, W. P., Bekarkhanechi, F. M., Poisson, H., Broussard, C., Ukwu, G., Auguste, E., Nguyen, D. D., Sehabi, G., Furyk, S. E., Gedeon, A. P., Onesis, O., El Aouame, A. M., Khodabocus, S. N., Shah, M., & Labelle, P. R. (2022). The global evolution of mental health problems during the COVID-19 pandemic: A systematic review and meta-analysis of longitudinal studies. *Journal of Affective Disorders*, *315*, 70–95. https://doi.org/10.1016/j.jad.2022.07.011

Collins, L. M., Schafer, J. L., & Kam, C-M. (2001). A comparison of inclusive and restrictive strategies in modern missing data procedures. *Psychological Methods*, *6*(4), 330–351. https://doi.org/10.1037/1082-989X.6.4.330

Czeisler, MÉ, Wiley, J. F., Czeisler, C. A., Rajaratnam, S. M. W., & Howard, M. E. (2021). Uncovering survivorship bias in longitudinal mental health surveys during the COVID-19 pandemic. *Epidemiology and Psychiatric Sciences*, *30*, e45. https://doi.org/10.1017/S204579602100038X

Della Cattaneo Volta, M. F., Vallone, F., & Zurlo, M. C. (2024). COVID-19-related stressors and psychophysical health conditions among Italian university students: A post pandemic insight. *Healthcare*, *12*(7), 752. https://doi.org/10.3390/healthcare12070752

Di Quirico, R. (2023). Uncertainty, anxiety and the post-pandemic economic environment. *Clinical Neuropsychiatry*, *20*(4), 227–232. https://doi.org/10.36131/cnfioritieditore20230401

Donohoe-Bales, A., O'Dean, S., Smout, S., Boyle, J., Barrett, E., Teesson, M., & Bower, M. (2024). What set some young adults apart during the COVID-19 pandemic? Mental health trajectories, risk and protective factors in an Australian longitudinal study. *Australian & New Zealand Journal of Psychiatry*, *58*(5), 435–445. https://doi.org/10.1177/00048674231223690

Eisinga, R., Grotenhuis, M. t., & Pelzer, B. (2013). The reliability of a two-item scale: Pearson, Cronbach, or Spearman-Brown? *International Journal of Public Health*, *58*(4), 637–642. https://doi.org/10.1007/s00038-012-0416-3

Fu, R., & Xie, L. (2024). Do public health events promote the prevalence of adjustment disorder in college students? An example from the COVID-19 pandemic. *International Journal of Mental Health Promotion*, *26*(1), 21–30. https://doi.org/10.32604/ijmhp.2023.041730

Fukase, Y., Ichikura, K., & Tagaya, H. (2023). Symptoms and risk factors of depression and PTSD in the prolonged COVID-19 pandemic: A longitudinal survey conducted from 2020 to 2022 in Japan. *BMC Psychiatry*, *23*(1), 180. https://doi.org/10.1186/s12888-023-04670-7

Gianfredi, V., Mazziotta, F., Clerici, G., Astorri, E., Oliani, F., Cappellina, M., Catalini, A., Dell'Osso, B. M., Pregliasco, F. E., Castaldi, S., & Benatti, B. (2024). Climate change perception and mental health. Results from a systematic review of the literature. *European Journal of Investigation in Health, Psychology and Education*, *14*(1), 215–229. https://doi.org/10.3390/ejihpe14010014

Graham, J. W. (2009). Missing data analysis: Making it work in the real world. *Annual Review of Psychology*, *60*(1), 549–576. https://doi.org/10.1146/annurev.psych.58.110405.085530

Harris, B. E., Rice, K., Murray, C. V., & Thorsteinsson, E. B. (2023). Validation of the brief adjustment disorder new modules with Australian oncology patients. *BioPsychoSocial Medicine*, *17*(1), 2. https://doi.org/10.1186/s13030-022-00259-w

Harris, E. (2023). Who declares end of COVID-19 global health emergency. *JAMA*, *329*(21), 1817. https://doi.org/10.1001/jama.2023.8656

Hettich, N., Entringer, T. M., Kroeger, H., Schmidt, P., Tibubos, A. N., Braehler, E., & Beutel, M. E. (2022). Impact of the COVID-19 pandemic on depression, anxiety, loneliness, and satisfaction in the German general population: A longitudinal analysis. *Social Psychiatry and Psychiatric Epidemiology*, *57*(12), 2481–2490. https://doi.org/10.1007/s00127-022-02311-0

Hilal, S., Kolozsvári, L. R., Indrayathi, P. A., Saeed, S. N., & Rurik, I. (2024). Psychological distress and food insecurity among international students at a Hungarian university: A post-pandemic survey. *Nutrients*, *16*, 241. https://doi.org/10.3390/nu16020241

Hippel, P. T. v. (2020). How many imputations do you need? A two-stage calculation using a quadratic rule. *Sociological Methods & Research*, *49*(3), 699–718. https://doi.org/10.1177/0049124117747303

Hlynsson, J. I., & Carlbring, P. (2024). Diagnostic accuracy and clinical utility of the PHQ-2 and GAD-2: A comparison with long-format measures for depression and anxiety. *Frontiers in Psychology*, *15*, 1259997. https://doi.org/10.3389/fpsyg.2024.1259997

Hobfoll, S. E. (1989). Conservation of resources: A new attempt at conceptualizing stress. *American Psychologist*, *44*(3), 3. https://doi.org/10.1037/0003-066X.44.3.513

IBM Corp. (2021). *IBM SPSS statistics for windows, version 28.0 [computer software]*. IBM Corp.

Janoff-Bulman, R. (1992). *Shattered assumptions: Towards a new psychology of trauma*. Free Press.

Kałwak, W., Weziak-Bialowolska, D., Wendołowska, A., Bonarska, K., Sitnik-Warchulska, K., Bańbura, A., Czyżowska, D., Gruszka, A., Opoczyńska-Morasiewicz, M., & Izydorczyk, B. (2024). Young adults from disadvantaged groups experience more stress and deterioration in mental health associated with polycrisis. *Scientific Reports*, *14*(1), 8757. https://doi.org/10.1038/s41598-024-59325-8

Kazlauskas, E., Elklit, A., & Truskauskaite, I. (2022). A longitudinal course of ICD-11 adjustment disorder symptom profiles: A 12-month follow-Up study. *Psychopathology*, *55*(6), 373–381. https://doi.org/10.1159/000525630

Kazlauskas, E., Gegieckaite, G., Eimontas, J., Zelviene, P., & Maercker, A. (2018). A brief measure of the International Classification of Diseases-11 adjustment disorder: Investigation of psychometric properties in an adult help-seeking sample. *Psychopathology*, *51*(1), 10–15. https://doi.org/10.1159/000484415

Kenntemich, L., von Hülsen, L., Eggert, L., Kriston, L., Gallinat, J., Schäfer, I., & Lotzin, A. (2024). Trajectories of depressive and anxiety symptoms and associated risk factors during the COVID-19 pandemic in Germany: A longitudinal cohort study. *Journal of Affective Disorders*, *355*, 136–146. https://doi.org/10.1016/j.jad.2024.03.131

Kim, S-J. (2024). Explosive increase and decrease in articles, citations, impact factor, and immediacy index during the COVID-19 pandemic: A bibliometric study. *Science Editing*, *11*(2), 107–113. https://doi.org/10.6087/kcse.334

Knox, L., Karantzas, G. C., Romano, D., Feeney, J. A., & Simpson, J. A. (2022). One year on: What we have learned about the psychological effects of COVID-19 social restrictions: A meta-analysis. *Current Opinion in Psychology*, *46*, 101315. https://doi.org/10.1016/j.copsyc.2022.101315

Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2003). The Patient Health Questionnaire-2: Validity of a two-item depression screener. *Medical Care*, *41*(11), 1284–1292. https://doi.org/10.1097/01.MLR.0000093487.78664.3C

Lara-Cabrera, M. L., Betancort, M., Muñoz-Rubilar, A., Rodríguez-Novo, N., Bjerkeset, O., & Las Cuevas, C. d. (2022). Psychometric properties of the WHO-5 well-being index among nurses during the COVID-19 pandemic: A cross-sectional study in three countries. *International Journal of Environmental Research and Public Health*, *19*(16), 10106. https://doi.org/10.3390/ijerph191610106

Lawrance, E. L., Jennings, N., Kioupi, V., Thompson, R., Diffey, J., & Vercammen, A. (2022a). Psychological responses, mental health, and sense of agency for the dual challenges of climate change and the COVID-19 pandemic in young people in the UK: An online survey study. *The Lancet Planetary Health*, *6*(9), e726–e738. https://doi.org/10.1016/S2542-5196(22)00172-3

Lawrance, E. L., Thompson, R., Le Newberry Vay, J., Page, L., & Jennings, N. (2022b). The impact of climate change on mental health and emotional wellbeing: A narrative review of current evidence, and its implications. *International Review of Psychiatry (Abingdon, England)*, *34*(5), 443–498. https://doi.org/10.1080/09540261.2022.2128725

Lawrence, M., Homer-Dixon, T., Janzwood, S., Rockstöm, J., Renn, O., & Donges, J. F. (2024). Global polycrisis: The causal mechanisms of crisis entanglement. *Global Sustainability*, *7*, e6. https://doi.org/10.1017/sus.2024.1

Lawrence, M., Janzwood, S., & Homer-Dixon, T. (2022). *What is a global polycrisis. Version 2.0*. Cascade Institute. https://cascadeinstitute.org/technical-paper/what-is-a-global-polycrisis/.

Li, X., Wu, R., Wu, M-Y, & Zhu, G. (2024). Changes and predictors of mental health of Chinese university students after the COVID-19 pandemic: A two-year study. *Journal of Affective Disorders*, *352*, 1–9. https://doi.org/10.1016/j.jad.2024.02.037

Liu, D., Zhou, Y., Tao, X., Cheng, Y., & Tao, R. (2024a). Mental health symptoms and associated factors among primary healthcare workers in China during the post-pandemic era. *Frontiers in Public Health*, *12*, 1374667. https://doi.org/10.3389/fpubh.2024.1374667

Liu, H., Tao, T. J., Chan, S. K. Y., Ma, J. C. H., Lau, A. Y. T., Yeung, E. T. F., Hobfoll, S. E., & Hou, W. K. (2024b). Daily routine disruptions and psychiatric symptoms amid COVID-19: A systematic review and meta-analysis of data from 0.9 million individuals in 32 countries. *BMC Medicine*, *22*(1), 49. https://doi.org/10.1186/s12916-024-03253-x

Lotzin, A., Acquarini, E., Ajdukovic, D., Ardino, V., Böttche, M., Bondjers, K., Bragesjö, M., Dragan, M., Grajewski, P., Figueiredo-Braga, M., Gelezelyte, O., Javakhishvili, J. D., Kazlauskas, E., Knefel, M., Lueger-Schuster, B., Makhashvili, N., Mooren, T., Sales, L., Stevanovic, A., & Schäfer, I. (2020). Stressors, coping and symptoms of adjustment disorder in the course of the COVID-19 pandemic – Study protocol of the European Society for Traumatic Stress Studies (ESTSS) pan-European study. *European Journal of Psychotraumatology*, *11*(1), 1780832. https://doi.org/10.1080/20008198.2020.1780832

Lotzin, A., Krause, L., Acquarini, E., Ajdukovic, D., Ardino, V., Arnberg, F., Böttche, M., Bragesjö, M., Dragan, M., Figueiredo-Braga, M., Gelezelyte, O., Grajewski, P., Anastassiou-Hadjicharalambous, X., Javakhishvili, J. D., Kazlauskas, E., Lenferink, L., Lioupi, C., Lueger-Schuster, B., Tsiskarishvili, L., … ADJUST Study Consortium (2021). Risk and protective factors, stressors, and symptoms of adjustment disorder during the COVID-19 pandemic – First results of the ESTSS COVID-19 pan-European ADJUST study. *European Journal of Psychotraumatology*, *12*(1), 1–16. https://doi.org/10.1080/20008198.2021.1964197

Lu, L., Contrand, B., Dupuy, M., Ramiz, L., Sztal-Kutas, C., & Lagarde, E. (2022). Mental and physical health among the French population before and during the first and second COVID-19 lockdowns: Latent class trajectory analyses using longitudinal data. *Journal of Affective Disorders*, *309*, 95–104. https://doi.org/10.1016/j.jad.2022.04.095

Luo, M-M, Hao, M., Li, X-H, Liao, J., Wu, C-M, & Wang, Q. (2024). Prevalence of depressive tendencies among college students and the influence of attributional styles on depressive tendencies in the post-pandemic era. *Frontiers in Public Health*, *12*, 1326582. https://doi.org/10.3389/fpubh.2024.1326582

Mathieu, E., Ritchie, H., Rodés-Guirao, L., Appel, C., Giattino, C., Hasell, J., Macdonald, B., Dattani, S., Beltekian, D., Ortiz-Ospina, E., & Roser, M. (2020). Coronavirus Pandemic (COVID-19). https://ourworldindata.org/coronavirus.

Morgan, M. A., Spanovic Kelber, M., Workman, D. E., Beech, E. H., Garvey Wilson, A. L., Edwards-Stewart, A., Belsher, B. E., Evatt, D. P., Otto, J., Skopp, N. A., Bush, N. E., & Campbell, M. (2022). Adjustment disorders: A research gaps analysis. *Psychological Services*, *19*(2), 283–293. https://doi.org/10.1037/ser0000517

O'Connor, R. C., Wetherall, K., Cleare, S., McClelland, H., Melson, A. J., Niedzwiedz, C. L., O'Carroll, R. E., O'Connor, D. B., Platt, S., Scowcroft, E., Watson, B., Zortea, T., Ferguson, E., & Robb, K. A. (2021). Mental health and well-being during the COVID-19 pandemic: Longitudinal analyses of adults in the UK COVID-19 Mental Health & Wellbeing study. *The British Journal of Psychiatry*, *218*(6), 326–333. https://doi.org/10.1192/bjp.2020.212

Peck, J. A. (2021). The disproportionate impact of COVID-19 on women relative to men: A conservation of resources perspective. *Gender, Work & Organization*, *28*(S2), 484–497. https://doi.org/10.1111/gwao.12597

Pieh, C., Budimir, S., Humer, E., & Probst, T. (2021). Comparing mental health during the COVID-19 lockdown and 6 months after the lockdown in Austria: A longitudinal study. *Frontiers in Psychiatry*, *12*, 625973. https://doi.org/10.3389/fpsyt.2021.625973

R Core Team. (2022). *R: A language and environment for statistical computing [Computer software]*. R Foundation for Statistical Computing. https://www.R-project.org/.

Rzeszutek, M., & Gruszczyńska, E. (2023). Depression during the COVID-19 pandemic among people living with HIV: Are low HIV/AIDS stigma and high perceived emotional support protective resources? *Stress and Health: Journal of the International Society for the Investigation of Stress*, *39*(4), 884–893. https://doi.org/10.1002/smi.3231

Scharbert, J., Humberg, S., Kroencke, L., Reiter, T., Sakel, S., Horst, J. t., Utesch, K., Gosling, S. D., Harari, G., Matz, S. C., Schoedel, R., Stachl, C., Aguilar, N. M. A., Amante, D., Aquino, S. D., Bastias, F., Bornamanesh, A., Bracegirdle, C., Campos, L. A. M., … Back, M. D. (2024). Psychological well-being in Europe after the outbreak of war in Ukraine. *Nature Communications*, *15*(1), 1202. https://doi.org/10.1038/s41467-024-44693-6

Staples, L. G., Dear, B. F., Gandy, M., Fogliati, V., Fogliati, R., Karin, E., Nielssen, O., & Titov, N. (2019). Psychometric properties and clinical utility of brief measures of depression, anxiety, and general distress: The PHQ-2, GAD-2, and K-6. *General Hospital Psychiatry*, *56*, 13–18. https://doi.org/10.1016/j.genhosppsych.2018.11.003

Takaew, T., Wongrattanakamon, S., Srisookkum, T., Katkhaw, O., Chairinkam, S., & Lamluk, P. (2024). Predictive factors of stress among the population in Pong District, Phayao Province, Thailand, after the COVID-19 outbreak. *Journal of Health Research*, *38*(2), 170–176. https://doi.org/10.56808/2586-940X.1072

Thompson, N., Kyaw, K. W. Y., Singh, L., Cikomola, J. C., Singh, N. S., & Roberts, B. (2024). The effect of COVID-19 on the non-COVID health outcomes of crisis-affected peoples: A systematic review. *Conflict and Health*, *18*(1), 37. https://doi.org/10.1186/s13031-024-00592-7

Topp, C. W., Østergaard, S. D., Søndergaard, S., & Bech, P. (2015). The WHO-5 Well-Being Index: A systematic review of the literature. *Psychotherapy and Psychosomatics*, *84*(3), 167–176. https://doi.org/10.1159/000376585

White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, *30*(4), 377–399. https://doi.org/10.1002/sim.4067

Wirkner, J., & Brakemeier, E. (2024). The crisis is over, long live the crisis: Mental health in emerging adulthood during the course of the COVID-19 pandemic. *Frontiers in Psychology*, *15*, 1283919. https://doi.org/10.3389/fpsyg.2024.1283919

Witteveen, A. B., Young, S. Y., Cuijpers, P., Ayuso-Mateos, J. L., Barbui, C., Bertolini, F., Cabello, M., Cadorin, C., Downes, N., Franzoi, D., Gasior, M., Gray, B., Melchior, M., van Ommeren, M., Palantza, C., Purgato, M., van der Waerden, J., Wang, S., & Sijbrandij, M. (2023). COVID-19 and common mental health symptoms in the early phase of the pandemic: An umbrella review of the evidence. *PLOS Medicine*, *20*(4), e1004206. https://doi.org/10.1371/journal.pmed.1004206

World Bank. (2025, January 16). GDP per capita, PPP. https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD?end=2021&locations=AT-HR-DE-GR-PT&start=2021&view=bar.

World Health Organization. (1998). *Wellbeing measures in primary health care/the DEPCARE project*. WHO Regional Office for Europe. https://iris.who.int/bitstream/handle/10665/349766/WHO-EURO-1998-4234-43993-62027-eng.pdf?sequence=1&isAllowed=y.

World Health Organization. (2019). *The 11th Revision of the International Classification of Diseases*. World Health Organization. https://icd.who.int/browse11/l-m/en.

World Health Organization. (2020). WHO characterizes COVID-19 as a pandemic. https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen.

World Health Organization. (2024). Population based on various sources – with major processing by Our World in Data. https://ourworldindata.org/covid-jhu-who.

Wu, Y., Dai, Z., Jing, S., Liu, X., Zhang, L., Liu, X., Ren, T., Fu, J., Chen, X., Xiao, W., Wang, H., Huang, Y., Wang, W., Gu, X., Ma, L., Zhang, S., Yu, Y., Li, L., Han, Z., … Qiao, Y. (2024). Prevalence and influencing factors of PTSD symptoms among healthcare workers: A multicenter cross-sectional study during the surge period of the COVID-19 pandemic since December 2022 in the Chinese mainland. *Journal of Affective Disorders*, *348*, 70–77. https://doi.org/10.1016/j.jad.2023.12.008

Würtzen, H., Clausen, L. H., Andersen, P. B., Santini, Z. I., Erkmen, J., & Pedersen, H. F. (2022). Mental well-being, health, and locus of control in Danish adults before and during COVID-19. *Acta Neuropsychiatrica*, *34*(2), 93–98. https://doi.org/10.1017/neu.2021.37

Xin, M., Yang, C., Zhang, L., Gao, C., & Wang, S. (2024). The impact of perceived life stress and online social support on university students' mental health during the post-COVID era in Northwestern China: Gender-specific analysis. *BMC Public Health*, *24*(1), 467. https://doi.org/10.1186/s12889-024-17935-x

Yang, N., Ma, H., Huang, M., Lau, E. Y. Y., Fong, D. Y. T., Wang, M., Wang, P., Xu, S., Xu, J., Jiang, C., Luo, Y., & Meng, R. (2023). Measurement properties and optimal cutoff point of the WHO-5 Among Chinese healthcare students. *Psychology Research and Behavior Management*, *16*, 5141–5158. https://doi.org/10.2147/PRBM.S437219

Zelviene, P., Kazlauskas, E., & Maercker, A. (2020). Risk factors of ICD-11 adjustment disorder in the Lithuanian general population exposed to life stressors. *European Journal of Psychotraumatology*, *11*(1), 1708617. https://doi.org/10.1080/20008198.2019.1708617

Zrnić Novaković, I., Lueger-Schuster, B., Verginer, L., Bakić, H., Ajduković, D., Borges, C., Figueiredo-Braga, M., Javakhishvili, J. D., Tsiskarishvili, L., Dragan, M., Nagórka, N., Anastassiou-Hadjicharalambous, X., Lioupi, C., & Lotzin, A. (2022). You can't do anything about it, but you can make the best of it: A qualitative analysis of pandemic-related experiences in six European countries. *European Journal of Psychotraumatology*, *13*(1), 2065431. https://doi.org/10.1080/20008198.2022.2065431